



Andrea Panebianco

ID: AX12345673 **Nationality:** Italian **Date of birth:** 24/06/1996 **Place of birth:** Catania, Italy

Gender: Male **Phone:** (+39) 3891161004 **Home Phone:** (+39) 095981309

Email: andrea.panebianco@spidmail.it **Email:** andrea.panebianco@unipa.it

LinkedIn: www.linkedin.com/in/andrea-panebianco

Home: Via Vittorio Emanuele, 560, 95033 Biancavilla (Italy)

ABOUT MYSELF

I obtained my Bachelor's Degree in Electronic Engineering on March 20, 2019, and my Master's Degree in Telecommunication Engineering on October 26, 2022, both from the University of Catania, Italy. I am currently a PhD student in Information and Communication Technologies at the University of Palermo, Italy. My research is centered on underwater acoustic communications.

WORK EXPERIENCE

Auburn University – Auburn, United States

City: Auburn | **Country:** United States | **Website:** <https://www.auburn.edu/> | **Name of unit/department:** Department of Electrical and Computer Engineering - **Business/sector:** Education

Visiting Ph.D. Student

[15/01/2025 – 14/07/2025]

Visiting Ph.D. Student for the research task entitled "Age of Channel State Information for Adaptive Power Control and Modulation in Underwater Acoustic Communications".

University of Catania – Catania, Italy

Address: Piazza Università 2, 95131 Catania (Italy) | **Website:** <https://www.unict.it> | **Email:** protocollo@unict.it | **Name of unit/department:** Department of Electrical electronic and computer engineering - **Business/sector:** Education

Academic tutor

[01/03/2024 – 01/06/2024]

Academic tutoring for the course "IoT and Big Data Sensing, Compression, and Communication", which included:

- Providing valuable information and advice to students;
- Offering educational support during lessons;
- Conducting practical or laboratory sessions to apply the theoretical topics covered in class.

University of Catania – Catania, Italy

Address: Piazza Università 2, 95131 Catania (Italy) | **Website:** <https://www.unict.it> | **Email:** protocollo@unict.it | **Name of unit/department:** Department of Electrical electronic and computer engineering - **Business/sector:** Education

Academic tutor

[15/03/2023 – 15/12/2023]

Academic tutoring for the course "IoT and Big Data Sensing, Compression, and Communication", which included:

- Providing valuable information and advice to students;
- Offering educational support during lessons;
- Conducting practical or laboratory sessions to apply the theoretical topics covered in class.

University of Catania – Catania, Italy

Address: Piazza Università 2, 95131 Catania (Italy) | **Website:** <https://www.unict.it> | **Email:** protocollo@unict.it | **Business/sector:** Education

Research intern

[08/03/2021 – 11/06/2021]

Research internship related to the development of an integrated system combining a web application, a simulated Wake Up Radio system, and a Digital Twin to monitor and manage underwater infrastructure sensor data. The web application, built with HTML, CSS, and PHP, provides user registration, a control dashboard, and data storage in a dedicated database. The Wake Up Radio system, using two Raspberry Pi Model 3B+ devices, facilitates communication via MQTT, with one device handling user requests (Low Radio) and the other collecting and transmitting sensor data (High Radio). A Digital Twin was implemented to ensure system reliability, monitoring messages from the High Radio and storing them in a database, while another database is employed for user management.

 **University of Catania** – Catania, Italy

Address: Piazza Università 2, 95131 Catania (Italy) | **Website:** <https://www.unict.it> | **Email:** protocollo@unict.it | **Name of unit/department:** Department of Electrical electronic and computer engineering - **Business/sector:** Education

Research intern

[01/09/2020 – 28/02/2021]

Research internship for the project titled "Secure and Efficient Wireless Communications - Intelligent Communication Algorithms and Integrated System Development for Underwater Wireless Communication Environments" involved developing an Android application with Android Studio to manage and gather sensor data from underwater infrastructure. Communication relies on the MQTT protocol, ensuring efficient data exchange. A Digital Twin was developed to oversee messages from the underwater environment and securely store them in a database. Furthermore, LoRa technology was integrated to enable long-range data transmission, ensuring effective communication and reliable data storage.

EDUCATION AND TRAINING

PhD in Information and Communication Technologies

University of Palermo [01/11/2022 – Current]

Address: Piazza Marina, 61, 90133 Palermo (Italy) | **Website:** <https://www.unipa.it>

Master's Degree in Telecommunications Engineering

University of Catania [02/10/2019 – 26/10/2022]

Address: Piazza Università, 2, 95131 Catania (Italy) | **Website:** <https://www.unict.it/> | **Field(s) of study:** Information and Communication Technologies (ICTs): • Inter-disciplinary programmes and qualifications involving Information and Communication Technologies (ICTs) | **Final grade:** 104/110 | **Level in EQF:** EQF level 6 | **National classification:** Master degree | **Type of credits:** ECTS | **Number of credits:** 120 | **Thesis:** Characterization of underwater communication channels in shallow water environments: from on-field measurements to Markov-based models

- Telecommunication Networks;
- Signal processing for multimedia applications;
- Multimedia Forensics;
- Antennas and Radiopropagation;
- Electronic for Telecommunications;
- Programming Techniques for distributed systems;
- Mobile Radio Networks;
- Internet;
- Microwave engineering;
- Design of telecommunications networks and systems;
- Protocols and architectures for the Internet of Things.

Bachelor Degree in Electronic Engineering

University of Catania [28/09/2015 – 20/03/2020]

Address: Piazza Università, 2, 95131 Catania (Italy) | **Website:** <https://www.unict.it/> | **Field(s) of study:** Information and Communication Technologies (ICTs): • Inter-disciplinary programmes and qualifications involving Information and Communication Technologies (ICTs) | **Final grade:** 96/110 | **Level in EQF:** EQF level 6 | **National classification:** Bachelor degree | **Type of credits:** ECTS | **Number of credits:** 180 | **Thesis:** An haptic interface for Tactile Internet scenario in 5G Networks

- Mathematical analysis 1, 2, 3;
- Linear algebra and geometry;
- Technical physics;
- Physics 1, 2;
- Inorganic chemistry;

- Economics applied to engineering;
- Business startups and business models;
- Electrical engineering;
- Electronics 1, 2;
- Systems theory;
- Automation controls;
- Signal theory;
- Telecommunications Fundamentals;
- Foundations of Informatics;
- Electronic calculators;
- Electronic measurements.

Scientific high school diploma

Liceo Ginnasio Statale "Giovanni Verga" [16/09/2010 – 07/07/2015]

Address: Via Salvo D'Acquisto, 95031 Adrano (Italy) | **Website:** <https://www.liceovergadrano.edu.it/> | **Final grade:** 110/110 with Laude | **Level in EQF:** EQF level 4 | **National classification:** High School Diploma

- Language and literature (Italian, English, Latin);
- Philosophy;
- History;
- Natural sciences and chemistry;
- Physics;
- Mathematics;
- Drawing and art history;
- Geography.

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING C2 READING C2 WRITING C2

SPOKEN PRODUCTION C2 SPOKEN INTERACTION C2

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user

SKILLS

Programming

Java language / IntelliJ IDEA CE / Python language / JavaScript language / PHP language / Message Queue technology (MQTT) / CSS language / Android Studio / C language / Matlab/SIMULINK / Relational databases (MySQL)

Mobile Forensics

Ares / Cellebrite / Nuance Forensics

Computer essentials

Windows / MacOS / Microsoft Office

Mobile

Android / iOS

PUBLICATIONS

[2025]

A Distributed Multi-Armed Bandit Approach for Modulation Adaptation in Underwater Networks In this work, we introduce a Multi-Player Multi-Armed Bandit (MP-MAB) framework for smart modulation adaptation in UnderWater Acoustic (UWA) Networks. Our solution is specifically tailored to run on resource-constrained UW nodes, thanks to the simplicity and low-

complexity of MAB. Our framework can leverage real-time throughput statistics to dynamically select the optimal modulation technique for multi-hop signal transmission in UW scenarios. Notably, this happens in a fully-distributed way on a per-node basis, as each UW node runs a local MAB agent to autonomously select the best modulation to use according to its own channel conditions. Using the DESERT UW simulator, we evaluate the performance of our proposed framework and compare it with alternative state-of-the-art learning approaches. Results demonstrate the higher efficiency and responsiveness of our algorithm compared to the alternatives, despite its simplicity and fully-decentralized nature.

WCNC 2025 - IEEE Wireless Communications and Networking Conference

[2025]

[MAGELLAN: A distributed MAB-based algorithm for Energy-Fair and Reliable Routing in Multi-Hop LoRa networks](#) In this work, we introduce MAGELLAN, a novel routing algorithm for multi-hop LoRa networks based on Multi-Armed Bandit (MAB) learning. MAGELLAN aims to achieve efficient packet delivery while minimizing the number of hops and fairly distributing the energy consumption across the network. We have conducted an extensive numerical evaluation by using the LoRaEnergySim simulator, and compared MAGELLAN against various routing approaches, including Leenders Multi-Hop Protocol (LMHP), a state-of-the-art routing algorithm. The results show that MAGELLAN achieves a superior performance in terms of energy-fairness and Packet Delivery Ratio (PDR), therefore improving the network lifetime and performance. More in detail, MAGELLAN achieves a reduction of up to 10% in the energy consumption as compared to a the random approach, and of to 14% in the PDR as compared to LMHP approach.

GLOBECOM 2024 - 2024 IEEE Global Communications Conference

[2025]

[AMUSE: a Multi-Armed Bandit Framework for Energy-Efficient Modulation Adaptation in Underwater Acoustic Networks](#) In this paper we present an enhanced version of AMUSE, a scalable and efficient framework designed to leverage the MAB approach for dynamic modulation selection, while enabling the optimization of various key performance metrics. Specifically, to illustrate the high level of flexibility of AMUSE in addressing multi-objective optimization, we here focus on the trade-off of Packet Error Rate (PER) and energy consumption across changing conditions, so as to make both reliability and energy efficiency the basis of the modulation adaptation decision-making process. Through extensive simulation in the DESERT simulator, we evaluate AMUSE performance against other state-of-the-art algorithms based on Deep Reinforcement Learning (DRL). Despite its simple design, AMUSE proves to be more efficient and responsive than the baselines, making it a powerful solution for improving UW communication performance. The results show that, in spite of the lightweight nature of AMUSE, our framework is able to outperform the DRL baselines by achieving an improvement of up to 23.64% in the network PER, and up to 80.65% in energy saving.

IEEE Open Journal of the Communications Society

[2024]

[Adaptive versus predictive techniques in underwater acoustic communication networks](#) Underwater communications suffer from numerous challenges typically associated with relevant signal attenuation, long propagation delay, limited available bandwidth, and high error rates that severely affect underwater transmission performance. Therefore, it is crucial to apply adaptive or predictive techniques to ensure the best possible performance and guarantee reliability in underwater communication, especially in rapidly changing environments. Using adaptive (i.e., reactive) or predictive (i.e., proactive) methods, it is possible to avoid data retransmission, improve the lifetime of underwater nodes, reduce maintenance frequency and the necessary equipment replacement and recharge, and consequently optimize performance in general. In this regard, many works in the literature propose various adaptive or predictive techniques for UnderWater Acoustic (UWA) networks, which we critically classify and discuss in this qualitative survey.

Elsevier - Computer Networks

[2024]

[Adaptive Modulation in Underwater Acoustic Networks \(AMUSE\): A Multi-Armed Bandit Approach](#) In this paper we propose AMUSE, the first Multi-Armed Bandit-based algorithm for smart modulation adaptation in Underwater Acoustic Networks. Note that AMUSE is specifically designed to suit resource-constrained UW nodes thanks to its simplicity and low-complexity. In particular, AMUSE relies on the current Packet Delivery Ratio (PDR) statistics to select in real time the best modulation technique to use for multihop signal transmission in the aforementioned UW scenarios. By employing the DESERT simulator we compare the performance achieved using AMUSE to those obtained using alternative state-of-the-art learning approaches. The results show that, in spite of its simplicity, our algorithm is more efficient and responsive than the other considered approaches.

[2024]

Balancing Optimization for Underwater Network Cost Effectiveness (BOUNCE): a Multi-Armed Bandit Solution In this paper we propose BOUNCE, a novel routing algorithm for UnderWater Acoustic (UWA) networks based on Multi-Armed Bandit. The main goal of BOUNCE is to efficiently route packets to ensure the highest transmission quality, the least packet latency, all while keeping an eye on a fair-balancing of the network energy consumption. We run an extensive simulation campaign, and tested BOUNCE against two baselines, namely the state-of-the-art RLOR algorithm, and a random approach. The results demonstrated how BOUNCE is able to outperform the other baselines in terms of energy consumption and fairness, packet latency, and Packet Delivery Rate. More in detail, BOUNCE achieves a reduction of up to 9.5% in the energy consumption and of up to 41% in the network latency, as well as an improvement of up to 27.5% in the Packet Delivery Ratio.

2024 IEEE International Conference on Communications Workshops (ICC Workshops)

[2024]

A comparative analysis of predictive channel models for real shallow water environments In this paper, we compare the performance of different types of SNR-based predictive models (i.e., Markov models, Hidden Markov models) in terms of balance between accuracy and complexity. We also provide a Kalman filter-based prediction of SNR values and compare this prediction with the performance achieved with the Markov models above mentioned. The models we have considered to carry out the comparison analysis have been developed based on real shallow water traces taken over the Tyrrhenian Sea, Italy.

Elsevier - Computer Networks

[2023]

Network intelligence vs. jamming in underwater networks: how learning can cope with misbehavior In this paper, we present a machine-learning technique to counteract jamming attacks in underwater networks. Indeed, this is relevant in security applications where sensor devices are located in critical regions, for example, in the case of national border surveillance or for identifying any unauthorized intrusion. To this aim, a multi-hop routing protocol that relies on the exploitation of a Q-learning methodology is presented with a focus on increasing reliability in data communication and network lifetime. Performance results assess the effectiveness of the proposed solution as compared to other efficient state-of-the-art approaches.

Frontiers in Communications and Networks

[2023]

Underwater Acoustic Channel Models for SNR Prediction in a Real Shallow Water Environment In this paper, we derive an underwater acoustic channel model based on field test measurements carried out in the Tyrrhenian sea. Specifically two models are obtained, one based on a Markov channel modeling with multiple possible states and another based on Kalman filter design. We compare their accuracy in predicting channel status and show that, based on the specific channel setting and application scenario, different channel models can be chosen to identify a trade-off between good prediction accuracy and affordable complexity.

2023 IEEE International Conference on Communications Workshops (ICC Workshops)

[2022]

An integrated acoustic/LoRa system for transmission of multimedia sensor data over an Internet of Underwater Things In this paper, we report about the design and development of an integrated acoustic/LoRa system for transmission of multimedia sensor data over the Internet of Underwater Things. We detail the blocks composing the system and show results obtained through field tests that assess the possibility to transmit multimedia data, as well as control data, for real time reconfiguration of the system parameters through properly designed Android or Web applications.

Elsevier - Computer Communications

PROJECTS

[01/11/2022 – Current]

AQUASMARTT This project aims at developing sustainable and accessible sensor networks for coastal environment monitoring, integrating both surface and underwater technologies. My contribution focuses on studying and applying predictive and adaptive techniques for underwater wireless networks (UWAN) using machine learning to optimize communications in complex and dynamic environments. I am working on designing intelligent models capable of analyzing real-time data, predicting environmental conditions, and dynamically adapting network strategies to improve energy efficiency and transmission reliability.

This approach is crucial to addressing the unique challenges of underwater networks and ensuring the sustainability of the developed solutions.

[01/07/2023 – 31/10/2023]

4Frailty This project focuses on improving the quality of life for frail individuals by developing innovative technological solutions for remote monitoring and care outside of hospital settings. This project integrates non-invasive systems, such as wearable sensors and environmental sensors, to track health parameters like oxygen levels, heart rate, and body temperature. These data are transmitted to healthcare professionals through a system that allows them to monitor patients in real time, detect risks early, and provide timely, low-impact interventions. To support this, an Android application was developed using Android Studio, enabling doctors to remotely monitor and manage patients' health. Through the app, healthcare providers can select a patient from their list and view medical parameters collected from biomedical sensors. The system uses the MQTT protocol for reliable communication between devices. To ensure system reliability and efficient data management, LoRa technology is used for long-distance communication and efficient storage in a healthcare database.

[01/05/2022 – 11/10/2022]

Characterization of underwater communication channels in shallow water environments: from on-field measurements to Markov-based models This Master's thesis is focused on developing an acoustic channel model for shallow water environments. This channel differs from typical radio channels, as it has a limited bandwidth that varies with frequency and propagation. Acoustic signals within this bandwidth are affected by time-varying multipath, which causes symbol interference and Doppler shifts. To characterize the channel, I used recorded traces from the La Spezia port area, provided by NATO's CMRE. Several finite-state Markov models were considered to evaluate the packet error rate, accounting for quality variations in the transmission of underwater acoustic signals. The entire system was implemented using MATLAB.

[01/04/2021 – 18/05/2021]

Derivation of a Wake Up Radio System This project involves the development of a comprehensive system combining a web application, a simulated Wake Up Radio system, and a Digital Twin to monitor and manage underwater infrastructure sensor data. The web application, built using Visual Studio Code, features a front-end designed with HTML and CSS and a back-end implemented with PHP. This application allows users to register and access a dashboard for system control, while also managing the storage of sensor data in a dedicated database. The simulated Wake Up Radio system uses two Raspberry Pi Model 3B+ devices connected via a serial port. One Raspberry Pi acts as the Low Radio, responsible for receiving requests from users, while the other functions as the High Radio, collecting sensor data and transmitting it to the user. Communication between components is handled through the MQTT protocol. A Digital Twin was implemented to ensure system reliability, monitoring messages from the High Radio and storing them in a database, while another database is employed for user management.

[01/09/2020 – 01/02/2021]

Secure and Efficient Wireless Communications - Intelligent Communication Algorithms and Integrated System Development for Underwater Wireless Communication Environments This project focuses on developing an Android application using the Android Studio programming environment. The application is designed to control and collect sensor data from underwater infrastructure, using the MQTT protocol for communication. To ensure the system operates correctly, a Digital Twin was implemented to monitor messages from the underwater environment and store them in a database. Additionally, data is transmitted using LoRa technology, enabling long-distance communication and efficient data storage in the database.

[01/06/2020 – 16/07/2020]

GitFlock, the java-based social network This project aims to create a chat system called GitFlock, which allows users to connect and start a chat, similar to a mailbox. To begin, the client must enter the server's address and port. The user can then choose between two options: the first requires a login session, where the user must provide their first name, last name, age, and a username to use on the platform. Access is restricted to users aged 14 and above. The system uses a concurrent hashmap to store connected users and removes them when they disconnect. The disconnection does not close the app, allowing the user to log in again with different credentials. The second option provides information about the app and instructions on how to use it. Within the app, there is a submenu with several options. The first option allows the user to view a list of online users, which corresponds to the hashmap of connected users. The second option lets the user check messages received during the session and send messages to other users by entering their username. Messages can be sent until "EXIT" is entered. A user can only contact another user who is online. Finally, the third option in the submenu allows the user to disconnect from the system, removing them from the hashmap and marking them as "unavailable" to other users. After disconnection, the user is returned to the main menu.

Link: <https://github.com/andreapanebianco/UniversityProject>

[01/06/2019 – 30/09/2020]

An haptic interface for Tactile Internet scenario in 5G Networks This Bachelor's thesis focuses on the study of the construction and use of the Hapkit Device in collaboration with Saint Louis University. The device is controlled through a 5G network and an application on an external terminal. The reception speed of instructions by the device was analyzed, and the application of the device in handling different physical scenarios through a Mininet virtual network was explored. Measurements were taken to assess the device's delay and response time under varying levels of introduced delay.

[01/09/2013 – 07/07/2015]

The Multilateral Comenius Project I participated in the European project "The Multilateral Comenius Project" LLP n°2013-1-DE3-COMO6-35867-3, where I traveled to Estonia to conduct comparative research between Italy and Estonia, working as part of a team with students from various European countries.

[25/03/2025]

DigComp 2.2 Certificate for Qualified Computer User

Official certification issued by AICA and accredited by Accredia, aligned with the European DigComp 2.2 framework. It certifies the candidate's digital competence across five key areas:

- Information and data literacy;
- Communication and collaboration;
- Digital content creation;
- Safety;
- Problem solving.

[23/11/2023]

Information Engineering License

Examination for the qualification to practice the profession of Information Engineer passed with 96/100.

[07/05/2022]

BI Level C2 Certificate in ESOL International (C2 CEFR)

Certificate, with identification number BIEB-0104082, certifying my C2 level knowledge of the English language.

[17/04/2020 – 30/04/2020]

Cisco Networking Academy " Introduzione a Internet of Everything" Certificate

Certificate suitable to:

- understand impact, benefits and challenges of the Internet of Everything;
- understand characteristics of the interaction between people, processes, data and objects that constitute the Internet of Everything;
- understand ways of connecting IP objects and non-IP objects to an object network for communication in the IoT environment;
- create models and prototypes in IoT.

[01/04/2020 – 16/04/2020]

Cisco Networking Academy "Introduction to Cybersecurity Cisco Networking Academy" Certificate

Certificate suitable to:

- know global implications of cyber threats;
- understand ways in which networks are vulnerable to attack;
- know impact of cyber attacks on industries;
- understand Cisco's approach to threat detection and defense;
- know why cybersecurity is a growing profession;
- know opportunities available to pursue network security certifications.

[01/04/2020 – 14/04/2020]

Cisco Networking Academy "Get Connected" Certificate

Certificate suitable to:

- identify different types of computer systems, internal components, and external devices;
- understand the directory structure in Microsoft Windows, and how to work with files and folders using a text editor;
- understand computer networks, browsers, Internet search, and email use;
- create and use various types of social media accounts including Facebook, LinkedIn and YouTube;
- identify the most common problems and implement simple solutions for hardware, software and networks.

[01/12/2016 – 08/12/2016]

Google Digital Training "Fundamentals of Digital Marketing" Certificate

Certificate in online management and advertising for businesses using specific communication channels like social networks.

[12/01/2015]

ECDL Certificate - Full Standard

Certificate, with identification number IT2056857, suitable to:

- understand the fundamental concepts relating to ICT (Information and Communication Technologies), computers, electronic devices and software;
- turn a computer on and off;
- operate effectively on a computer desktop using icons and windows;
- adjust key system settings and use online help features;
- create a simple document and print copies;
- know the main file management concepts and be able to effectively organize folders and files;
- operate adequately with applications relating to the Microsoft Office suite;
- understand the fundamental concepts relating to storage media and the use of software for compression and extraction of large files;
- understand concepts relating to networks and connection possibilities, and be able to connect to a network;
- understand the importance of making backup copies of data and protecting data and electronic devices from malware;
- understand the importance of "green computing", accessibility and safeguarding users' health.

[04/02/2015]

Cambridge English Level 1 Certificate in ESOL International (First)

Certificate, with identification number 14CIT0060020, certifying my B2 level knowledge of the English language.

[10/11/2014]

ECDL IT-Security Certificate - Specialised level

Certificate, with identification number IT2056857, suitable to:

- understand the concepts relating to network navigation and information security;
- use the web browser and manage its settings, bookmarks/favorites and printing of web pages;
- carry out information searches on the internet effectively and evaluate the content of the web critically;
- understand the main issues related to copyright and data protection;
- understand the concepts of social networks, communications and email;
- send and receive email messages and manage the settings of an email program;
- organize and search email messages;
- use calendars.

[30/09/2014]

Certification Tõend

Certificate issued by the "Tallinn Technical Secondary School" confirming my participation to the European project "The Multilateral Comenius Project" traveling to Estonia from 24/09/2014 to 30/09/2014 to conduct research with a team of students from various European countries.

[15/10/2010]

ECDL Certificate

Certificate with identification number MED003134, suitable to:

- understand the fundamental concepts relating to ICT (Information and Communication Technologies), computers, electronic devices and software;
- turn a computer on and off;
- operate effectively on a computer desktop using icons and windows;
- adjust key system settings and use online help features;
- create a simple document and print copies;
- know the main file management concepts and be able to effectively organize folders and files;
- understand the fundamental concepts relating to storage media and the use of software for compression and extraction of large files;
- understand concepts relating to networks and connection possibilities, and be able to connect to a network;
- understand the importance of making backup copies of data and protecting data and electronic devices from malware;
- understand the importance of "green computing", accessibility and safeguarding users' health.

HONOURS AND AWARDS

[12/09/2023] GTTI Association - Telecommunications and Information Technologies Group

2023 Francesco Carassa Award I was granted with this award during the 2023 GTTI Annual Meeting for presenting my research poster titled "Characterization of Underwater Acoustic Channel Models for SNR Prediction in Real Shallow Water Environments".

[18/12/2019] University of Catania

TLC for a better world: smart scenarios, technologies and ideas, Student Category: Innovative Prototype I received this recognition at TelcomDay 2019 for my project "An Haptic Interface for Tactile Internet Scenarios in 5G Networks".

[12/12/2014] Sicily Region

Sicilian Regional Award "The Positive" Values in Movement I was awarded with this prize by the Sicily Region, with the following recognition: "The student has demonstrated a productive study approach in all subjects, constructive participation in educational discussions, responsible behavior, and analytical understanding throughout their academic journey".

[23/05/2011] University of Catania

Etniade 2011 I was ranked in the mathematics competitions organized by the University of Catania, called Etniadi.

CONFERENCES & SEMINARS

[02/02/2024] University of Padua

A Multi-Armed Bandit approach in Underwater networks Seminar held at the Winter School on Underwater Network Simulations and Experimentation (UNWiS) of the University of Padua where I presented various artificial intelligence techniques, focusing on the Multi-Armed Bandit (MAB) applied to underwater networks, delving into a study I conducted in where MAB outperformed Deep Reinforcement Learning approaches, precisely in this context.

[28/05/2023 – 28/05/2023] Roma Convention Center "La Nuvola"

IEEE International Conference on Communications - WS-24: 1st Workshop on Underwater Communication, Access and Networking (UCAN) Presentation of my research work: "Underwater Acoustic Channel Models for Predicting Signal-to-Noise Ratio in a Real Shallow Water Environment".

Link: <https://icc2023.ieee-icc.org/program/workshops#ws-24>

[18/05/2021] Università degli Studi di Catania

Development and Applications of Innovative Materials and Processes for the Diagnostics and Restoration of Cultural Heritage Seminar where I presented the project related to the paper "Design and experimentation of an integrated acoustic/LoRa system for transmission of multimedia sensor data over an Internet of Underwater Things" at the seminar, as well as the project "Derivation of a Wake Up Radio System."

ATTENDED SEMINARS

[25/01/2024 – 08/03/2024]

Advanced modeling, design and protocol solutions for communication networks

Seminars covering the following topics:

- Basics on stochastic processes;
- Markov processes;
- Discrete Time Markov Chains;
- Continuous Time Markov Chains;
- Kendall notation;
- Arrival process;
- Service process;
- M/M/1 queue;
- M/M/m queue;
- Comparison between different queue types.

[14/11/2022 – 22/11/2022]

Emerging Network Technologies

Seminars presenting some emerging trends in network technologies, particularly open architectures for beyond 5G cellular systems and solutions for mass IoT applications. It also addressed some cutting-edge topics, such as methodologies for zero-touch networks beyond 5G and quantum information.

[04/11/2019 – 09/12/2019]

From soft skills to the job market

Seminars regarding how to: adequately create a curriculum vitae, have a job interview, use social networks to search for a job.

[26/11/2019]

Digital Forensics - Computer forensics

Seminar on how to extract data from corrupted files.

[20/11/2019]

Mobile Forensics - Analysis and processing of extracted data

Seminar regarding operational best practices with references to regulations and investigative methodologies in the field of Mobile Forensics.

[19/11/2019]

Mobile Forensics - Procedures for data extraction

Seminar on data extraction methods from devices (Chip Off, JTAG & Cloud) in the field of Mobile Forensics.

[18/04/2018 – 24/05/2018]

Electronics in cars

Seminars organized by STMicroelectronics focusing on the study of intelligent security systems in vehicles, with particular emphasis on the development and application of radar technologies.

MANAGEMENT AND LEADERSHIP SKILLS

Ability to work in stressful situations

Ability to work in stressful situations, both as part of a team and independently, related to fiscal deadlines for project deliveries; with a focus on punctuality and adherence to proposed guidelines.

COMMUNICATION AND INTERPERSONAL SKILLS

Ability to work in teams

Ability to work in teams experienced in multiple situations, such as academic and research projects.

DRIVING LICENCE

Driving Licence: AM

Driving Licence: B

HOBBIES AND INTERESTS

Technology lover I really enjoy keeping myself up to date on new technologies, trying to work with them and experiment with them in order to gain experience.

Nature lover I really enjoy hiking and going on excursions, both on foot and by Mountain Bike, either alone or with company.

Swimming lover I have been swimming for several years, I consider it a relaxing and complete sport that trains every part of the body.

According to law 679/2016 of the Regulation of the European Parliament of 27th April 2016, I hereby express my consent to process and use my data provided in this CV and application for recruiting purposes.

Catania, 18/09/2025



Andrea Panebianco